

1.2: GAUSSIAN ELIMINATION

The objective of this section is to introduce an algorithm to solve any system of linear equations. This will be done in two steps:

- 1) Use (elementary) row operations to bring the augmented matrix to a "nice" form.
- 2) Solve the system corresponding to the "nice" augmented matrix.

Let's start by describing what a "nice" augmented matrix is for us:

DEF: A matrix A is in **row-echelon form** (and we say that it is a **row-echelon matrix**) if it satisfies the following three properties:

- 1) All zero rows (if any) are at the bottom.
- 2) For each nonzero row, its first nonzero element is a 1 (i.e. the row looks like $[0 \ 0 \ \dots \ 0 \ 1 \ * \ \dots \ *]$).

We call this first 1 a **leading 1**.

- 3) Every leading 1 is to the right of all leading 1's in the rows above it.

Moreover, we say that A is in **reduced row-echelon form** (and is a **reduced row-echelon matrix**) if it also satisfies

- 4) Each leading 1 is the only nonzero entry in its column.

EXAMPLES: The following matrices are in row-echelon form:

$$\begin{bmatrix} 1 & * \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & * & * & * & * \\ 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 1 & * \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The leading 1's (highlighted) form a "staircase" (echelon) pattern.

Moreover, the following matrices are in reduced row-echelon form:

$$\begin{bmatrix} 1 & * \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & * & 0 & 0 & * \\ 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 1 & * \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 & 0 & * & 0 & * & * \\ 0 & 0 & 0 & 1 & * & 0 & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

THEOREM: Every matrix can be brought to (reduced) row echelon form via elementary row operations.

To see why this is true, we will give an algorithm to calculate a (reduced) row-echelon form of a given matrix:

GAUSSIAN ALGORITHM: Suppose we are given any matrix, for instance:

$$A = \begin{bmatrix} 0 & 0 & 2 & -1 & 1 & 2 \\ 0 & 3 & 1 & 10 & 5 & 1 \\ 0 & 1 & 0 & 3 & 1 & -2 \end{bmatrix}$$

STEP 1: If A only consists of zeros, then we are done (it is already in row-echelon form). If not, continue

STEP 2: Find the leftmost column that is not zero, and use row operations to get a 1 at the top of this column:

$$\begin{bmatrix} 0 & 0 & 2 & -1 & 1 & 2 \\ 0 & 3 & 1 & 10 & 5 & 1 \\ 0 & 1 & 0 & 3 & 1 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \end{bmatrix}$$

First nonzero column Now 1 is on top

STEP 3: Use operations of type 3 to make every entry below the leading 1 a zero.

$$\begin{bmatrix} 0 & 1 & 0 & 3 & 1 & -2 \\ 0 & 3 & 1 & 10 & 5 & 1 \\ 0 & 0 & 2 & -1 & 1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 1 & 0 & 3 & 1 & -2 \\ & & & & & \\ & & & & & \end{bmatrix}$$

STEP 4: Repeat Steps 1-3 on the submatrix consisting of the remaining rows.

$$\begin{bmatrix} 0 & 1 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 1 & 2 & 7 \\ 0 & 0 & 2 & -1 & 1 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 1 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 1 & 2 & 7 \\ & & & & & \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 1 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 1 & 2 & 7 \\ & & & & & \end{bmatrix}$$

We arrived at a row-echelon matrix!

The algorithm finishes once we dealt with all nonzero rows, and the end result is a row-echelon matrix. If we want to obtain a reduced row-echelon matrix, we have to perform an extra step:

STEP 5: Using operations of type III, make every entry above the rightmost leading 1 a zero. Then do the same thing with the next leading 1 to the left, and so on.

$$\begin{bmatrix} 0 & 1 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 1 & 2 & 7 \\ 0 & 0 & 0 & 1 & 1 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 0 & 0 & 1 & 1 & 4 \end{bmatrix} \leftarrow \text{Reduced!}$$

↑
Rightmost leading 1

Now, how do we solve a linear system when its augmented matrix is in row-echelon form? There are two possibilities

CASE I: Suppose the augmented matrix has a row of the form $[0 \ 0 \ \dots \ 0 \ 0 \ | \ c]$ with $c \neq 0$.

Then the system contains an equation of the form $0 = c$, so the system is inconsistent

CASE II: Suppose the augmented matrix does not have a row of the form $[0 \ 0 \ \dots \ 0 \ 0 \ | \ c]$ with $c \neq 0$.

For example, consider the system

$$\begin{cases} x + 2y + z = 3 \\ y - z = 0 \end{cases} \quad \text{with augmented matrix} \quad \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

↑
Leading 1's

We call the variables corresponding to the leading 1's **leading variables**. To solve the system, we solve for the leading variables using the equations of the system, starting with the last equation and working our way up. The nonleading variables (if any) are set to be parameters.

$$\begin{cases} x + 2y + z = 3 \\ y - z = 0 \end{cases}$$

SOLUTION:

$x =$

$y =$

$z =$

NOTE: In the above example, we first had to solve for y in the second equation and use that information to solve for x in the first equation. This procedure is called **back-substitution** and is only necessary when the augmented matrix is not in reduced row-echelon form.

Let's put all of this to the test!

EXAMPLE: Solve the system

$$\begin{cases} 4x + y = 2 \\ 3x + 2y + 4z = 1 \\ 10x + 5y + 8z = 4 \end{cases}$$

SOL:

$$\left[\begin{array}{ccc|c} 4 & 1 & 0 & 2 \\ 3 & 2 & 4 & 1 \\ 10 & 5 & 8 & 4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} & & & \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} & & & \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} & & & \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} & & & \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} & & & \end{array} \right]$$

$$\left\{ \begin{array}{l} x = \\ y = \\ z = \end{array} \right. \Rightarrow \boxed{\begin{array}{l} x = \\ y = \\ z = \end{array}}$$

EXAMPLE (EXERCISE 1.1.12): Determine, in terms of the numbers a, b, c , when the following system is consistent:

$$\begin{cases} x + 2y - z = a \\ 2x + y + 3z = b \\ x - 4y + 9z = c \end{cases}$$

Find its solutions whenever it is consistent

Sol:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & a \\ 2 & 1 & 3 & b \\ 1 & -4 & 9 & c \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array} \right]$$
$$\rightarrow \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array} \right]$$

The system is consistent if and only if , i.e., if . Let's find its solutions in this case:

$$\left\{ \begin{array}{l} x = \\ y = \\ z = \end{array} \right.$$

RANK

In general, a matrix A will have more than one row-echelon form. However, one can show that:

- 1) The reduced row-echelon form of A is unique.
- 2) The number and position of the leading 1's in any row-echelon form of A (not only the reduced one) are always the same.

This motivates the following definition:

DEF: Let A be a matrix and B a row-echelon matrix such that A can be brought to B via elementary operations. We define the **rank of A** as

$$\text{rank } A = \text{n}^\circ \text{ of leading 1's in } B = \text{n}^\circ \text{ of nonzero rows in } B.$$

NOTE: $\text{rank } A$ is smaller than or equal to the number of rows of A , and it is also smaller than or equal to the number of columns of A .

EXAMPLE: Calculate the rank of

$$A = \begin{bmatrix} 1 & 1 & m & 1 \\ 1 & m & 1 & 1 \\ m & 1 & 1 & 1 \end{bmatrix}$$

in term of m

SOL:

$$\begin{bmatrix} 1 & 1 & m & 1 \\ 1 & m & 1 & 1 \\ m & 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix} \rightarrow \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}$$

If $m=1$, then A is transformed via row operations to

$$B = \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}, \text{ so rank } A =$$

If $m \neq 1$, we can divide by $m-1$:

$$\begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix} \rightarrow \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}$$

If $m = -2$, then A is transformed into

$$B = \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix} \rightarrow \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}, \text{ so rank } A =$$

If $m \neq 1, -2$, we can divide by $-m-2$:

$$\begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix} \rightarrow \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}, \text{ so rank } A =$$

CONCLUSION: rank $A =$ if $m=1$ and rank $A=3$ if $m \neq$.

THEOREM: Suppose a linear system of m equations in n variables is consistent and the rank of its augmented matrix is r . Then:

- 1) The set of solution involves exactly $n-r$ parameters.
- 2) If $r < n$, the system has infinitely many solutions.
- 3) If $r = n$, the system has exactly one solution.

PROOF :

- 1) From the Gaussian algorithm, we know that

$$n^{\circ} \text{ of parameter} = n^{\circ} \text{ of nonleading variables} = (n^{\circ} \text{ of variables}) - (n^{\circ} \text{ of leading variables}) = n - r$$

- 2) If $r < n$, then $n-r > 0$ and the solution has parameters, so there are infinitely many solutions.

- 3) If $r=n$, then $n-r=0$ and the solution does not have parameters, so there is only one solution. //

As a consequence, given a general system of linear equations, exactly three possibilities exist:

- 1) The system is inconsistent, so it has no solutions.
- 2) The system has exactly one solution.
- 3) The system has an infinite amount of solutions.